

Problem Statements

AI ML

1. **Predictive Supply Chain Routing using AI agents**

Global supply chains are increasingly vulnerable to disruptions caused by unpredictable factors such as extreme weather, geopolitical conflicts, infrastructure failures, and sudden demand shifts. Traditional logistics systems are largely reactive, relying on manual intervention after disruptions occur—often leading to cascading delays, increased costs, and operational inefficiencies.

This problem focuses on designing a scalable, AI-driven system that can continuously analyze diverse, real-time transit data to proactively detect potential disruptions and optimize routing decisions before they escalate.

Objective

Develop an intelligent platform that can:

Monitor and analyze multi-source data (traffic, weather, news, geopolitical signals (like wars), etc.) in real time

Predict and flag risks early, including unforeseen events like war or sudden route blockages

Dynamically recommend or execute optimized rerouting strategies to avoid bottlenecks

Minimize dependency on manual intervention through autonomous decision-making

Examples & Use Cases

If a conflict or war blocks a critical trade route, the system should immediately identify alternative paths and redistribute shipments. Similarly, weather disruptions or port congestion can be detected early, enabling preemptive rerouting to prevent large-scale delays.

AI Application

Leverage multiple specialized AI agents - such as news analysis agents, weather prediction models, and traffic intelligence systems - that collaborate to assess risks and make coordinated decisions. Use predictive modeling, reinforcement learning, and real-time optimization to continuously improve routing efficiency.

2. Open Router

With the rapid adoption of AI models, developers often face challenges in managing cost, latency, and efficiency when sending prompts to different LLM providers. Most systems rely on static routing or manual selection, leading to higher expenses, slower responses, and suboptimal resource utilization.

OpenRouter AI aims to build an intelligent routing layer that dynamically selects the most optimal model and endpoint for every request using AI/ML.

Objective

Develop a smart routing system that can:

Automatically route prompts to the most cost-efficient and performant model

Optimize for latency, token usage, and response quality

Detect and use the nearest or fastest available endpoint

Incentivize efficient usage by rewarding cost-saving behavior

Examples & Use Cases

A simple query can be routed to a low-cost model, while complex reasoning tasks are directed to more powerful models. High-traffic systems can dynamically balance load across providers to reduce latency and avoid rate limits.

AI Application

Use ML models to predict cost vs. quality trade-offs, classify prompt complexity, and continuously learn from usage patterns. Incorporate real-time analytics and feedback loops to improve routing decisions over time.

Innovation Opportunities

Build a self-learning router that improves with usage

Introduce a credit/reward system for efficient prompt usage

Enable fallback and failover intelligence across providers

Add privacy-aware routing and edge optimization

3. Kiosk Vision

Self-service kiosks are widely used across restaurants, retail stores, and pharmacies to drive sales and improve efficiency. However, these systems often fail to serve specially-abled

such as visual, hearing, or cognitive differences, limiting accessibility and overall user experience.

Kiosk Vision aims to build an AI-powered kiosk assistant that can make self-service systems inclusive for all users.

Objective

Develop an intelligent solution that can:

Assist specially-abled users through voice, visual, or gesture-based interaction

Provide real-time guidance by understanding on-screen content

Simplify ordering for users with low digital familiarity

Handling payments via cards, cash, etc. (a real bottleneck)

Examples & Use Cases

A visually impaired user can navigate menus using voice guidance, while a hearing-impaired user can rely on real-time text or visual cues. First-time users can be guided step-by-step with smart recommendations and simplified flows.

AI Application

Leverage AI/ML techniques such as computer vision for screen understanding, NLP for conversational assistance, and personalization models to adapt the kiosk experience based on user behavior and needs

3. Develop an AI system that detects and flags misinformation in real-time from live data streams such as social media or news feeds.

Cyber Security

1. AI-Powered Cybersecurity for Intelligent Industrial Systems

Industrial environments are increasingly adopting AI-driven machines, Industrial IoT (IIoT), and automated control systems to improve efficiency and decision-making. As Operational Technology (OT) networks become more connected with IT systems and cloud platforms, the risk of cyberattacks on critical machines and industrial processes is growing.

Develop an AI-powered cybersecurity solution that can monitor industrial OT networks, detect abnormal machine behavior, and identify potential cyber threats in real time without disrupting operations. The system should help secure communication between machines, control systems, and enterprise platforms while ensuring safety and operational continuity.

2. Modern data centers support critical digital infrastructure and must maintain high availability and reliability to ensure uninterrupted services. With increasing demand from cloud computing, AI workloads, and digital platforms, maintaining optimal performance of power, cooling, and IT infrastructure has become more complex.

Develop an AI-powered data center management solution that can monitor infrastructure systems such as power, cooling, and network equipment, predict potential failures, and optimize operations to maximize uptime. The system should use data analytics and machine learning to detect anomalies, recommend preventive actions, and ensure efficient and reliable data center operations.

3. Many organizations rely on periodic penetration tests to evaluate their security, but these assessments are often limited in scope and do not accurately replicate how modern attackers operate over time. Red teams need advanced infrastructure to simulate real-world adversary behavior, including scalable command-and-control (C2) management, stealthy payload delivery, multi-stage access, and coordinated control of compromised systems across different environments. However, existing tools are often complex, fragmented, or not designed for enterprise-scale simulations. The problem is the lack of a unified red-team platform that allows security professionals to efficiently deploy, manage, and operate large-scale C2 infrastructures and realistic attack scenarios to emulate advanced threat actors during authorized security assessments.

Robotics / IoT

Open innovation sticking to the theme.

Web3/Blockchain:

1.HUSH TRACK (<https://hushnetworks.in/hushchain>

2.)Decentralized Autonomous Supply Chain Finance (DeFi for B2B) Small and mid-sized suppliers in global supply chains are forced to wait 60–120 days for invoice payments from large buyers, creating severe cash flow crises. Traditional invoice financing through banks is slow, collateral-heavy, and excludes smaller players entirely. Meanwhile, large buyers have no incentive to pay early under existing systems. *Build a Web3-powered decentralized supply chain finance protocol where suppliers can tokenize unpaid invoices as on-chain assets, instantly access liquidity from a decentralized lending pool, and have repayments automatically triggered via smart contracts upon buyer payment confirmation — eliminating banks as intermediaries*

and giving every supplier in the chain fair access to working capital regardless of their size.

3. A specialization platform for doctors (some special field) to annotate images and get paid for annotation. The annotated is used for training a ml model. (make the platform).

Quantum Computing :

1. Quantum-Optimized Portfolio Risk Management for Financial Institutions Financial institutions manage thousands of assets simultaneously, requiring complex risk calculations — Value at Risk (VaR), stress testing, and portfolio rebalancing — that classical computers take hours to process. By the time results arrive, market conditions have already shifted, leading to delayed decisions and missed opportunities. *Build a quantum computing-powered financial risk engine that leverages quantum algorithms such as Quantum Monte Carlo and QAOA to evaluate millions of portfolio scenarios in real time — enabling banks, hedge funds, and asset managers to make faster, more accurate risk-adjusted investment decisions at a scale impossible for classical systems.*

2. Quantum-Enhanced Supply Chain Optimization for Global Enterprises Large enterprises manage supply chains with

thousands of variables — supplier locations, shipping routes, warehouse capacity, demand forecasts, and geopolitical constraints. Classical optimization algorithms fail to find the true global optimum, leaving businesses with inefficient routing, excess inventory, and higher logistics costs. Develop a quantum optimization platform using Variational Quantum Eigensolver (VQE) or quantum annealing to solve large-scale supply chain routing and inventory allocation problems — helping global businesses reduce logistics costs, minimize waste, and dynamically re-optimize supply chains in response to real-world disruptions.

3. Quantum-Safe Cryptography Migration Platform for Enterprise Security *As quantum computers advance, they will break widely used encryption standards like RSA and ECC, putting sensitive business data, financial transactions, and customer records at serious risk. Most enterprises have no roadmap to identify vulnerable systems or migrate to quantum-resistant cryptographic standards before the threat becomes real. Build an enterprise quantum-readiness platform that automatically audits an organization's entire cryptographic infrastructure, maps vulnerabilities to quantum threats, and orchestrates a phased migration to NIST-approved post-quantum cryptographic algorithms — ensuring businesses are protected before cryptographically relevant quantum computers arrive.*

